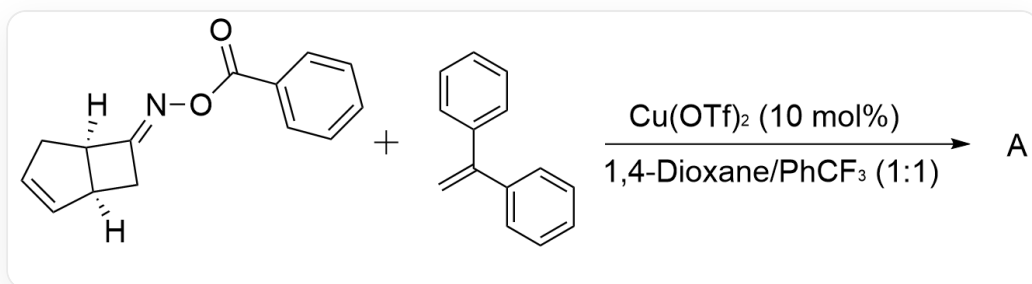


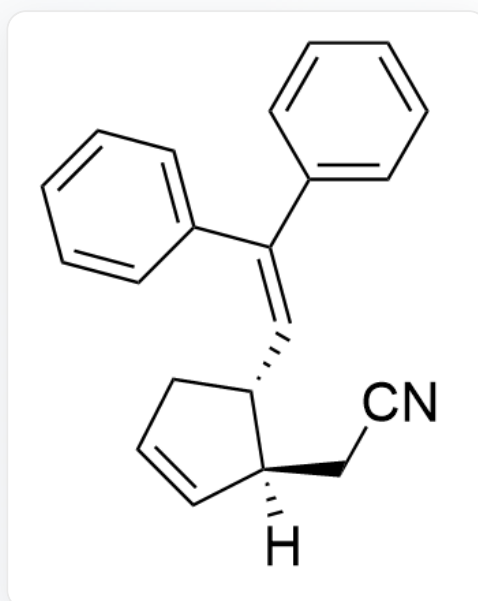
Question



[H][C@]12[C@](C/C2=NOC(C3=CC=CC=C3)=O)([H])C=CC1.C=C(C1=CC=CC=C1)C2=CC=CC=C2>Cu(OTf)2 (10 mol%), 1,4 - Dioxane/PhCF3 (1 : 1)>A

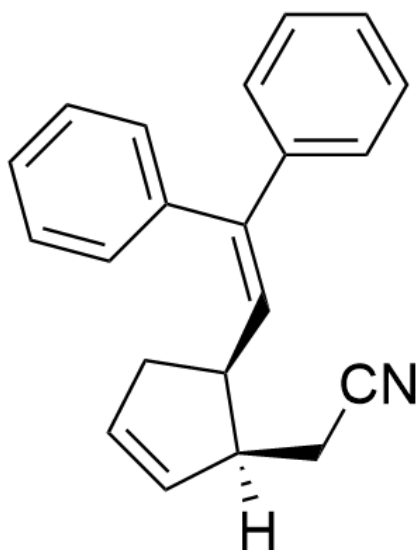
Please select the most suitable product A

A.



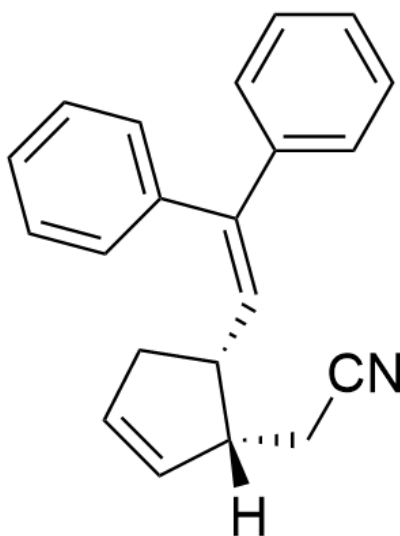
[H][C@]1(CC#N)C=CC[C@@H]1/C=C(C2=CC=CC=C2)/C3=CC=CC=C3

B.



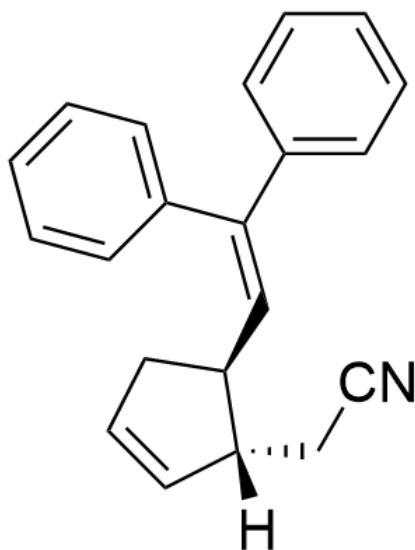
[H][C@]1(CC#N)C=CC[C@H]1/C=C(C2=CC=CC=C2)/C3=CC=CC=C3

C.



[H][C@@]1(CC#N)C=CC[C@@H]1/C=C(C2=CC=CC=C2)/C3=CC=CC=C3

D.



[H][C@@]1(CC#N)C=CC[C@H]1/C=C(C2=CC=CC=C2)/C3=CC=CC=C3

Answer

Correct Answer: **A**

Detailed Explanation

This reaction is a radical reaction

CHECKPOINT

1 PTS

This reaction is a radical reaction

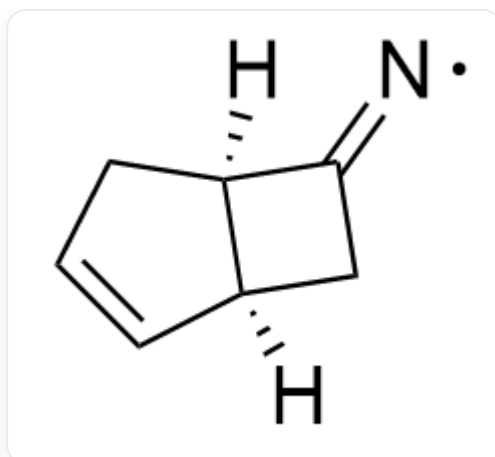
The Cu in the reaction system acts as a catalyst

CHECKPOINT

1 PTS

The Cu in the reaction system acts as a catalyst

$Cu(I)$ first donates a single electron, causing the cleavage of the weaker N-O bond and forming a radical intermediate



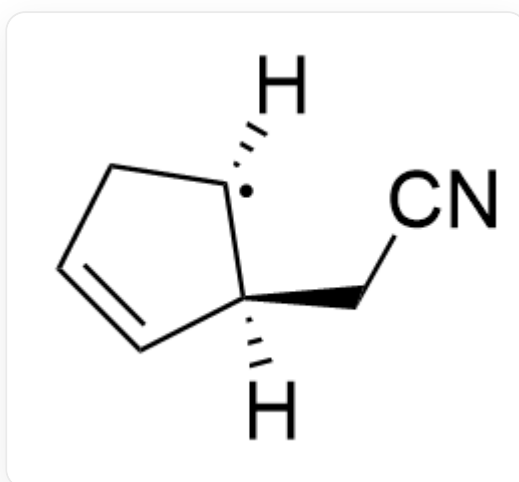
[H][C@]12[C@](CC2=[N])([H])C=CC1

CHECKPOINT

1 PTS

Cu(I) first donates a single electron, causing the cleavage of the weaker N-O bond and forming a radical intermediate

Subsequently, this unstable intermediate rapidly undergoes ring-opening to form a more stable carbon radical



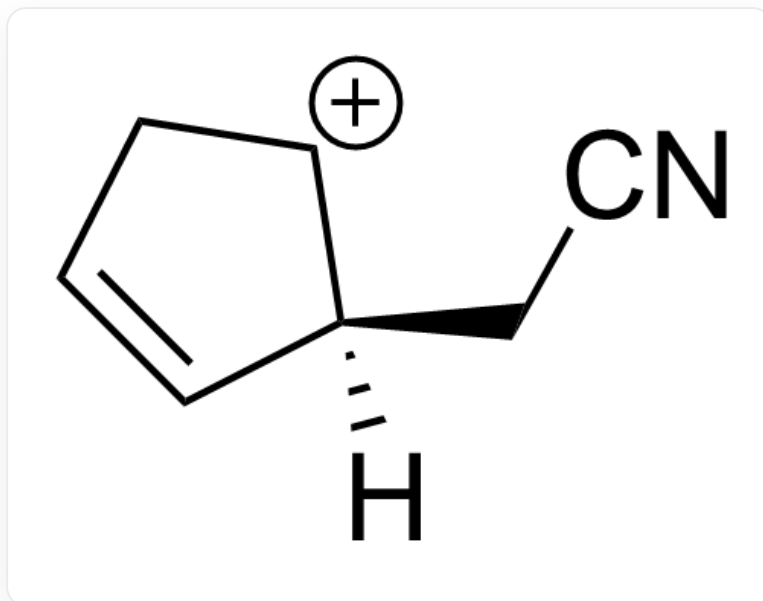
[H][C@]1(CC#N)C=CC[C1][H]

CHECKPOINT

1 PTS

Subsequently, this unstable intermediate rapidly undergoes ring-opening to form a more stable carbon radical

Next, $Cu(II)$ oxidizes this radical to form a carbocation



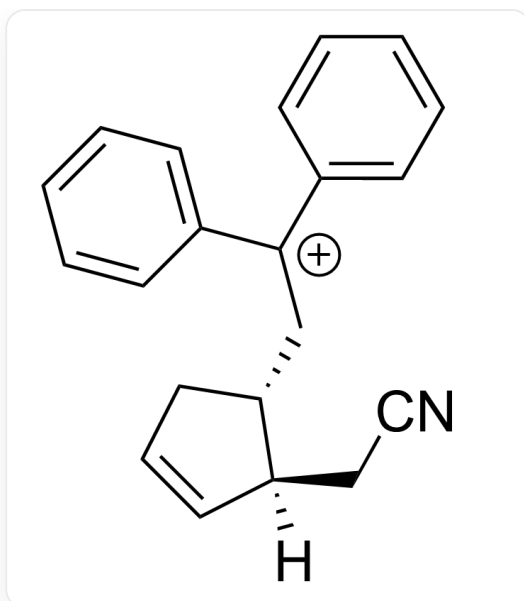
[H][C@]1(CC#N)C=CC[CH+]1

CHECKPOINT

1 PTS

Next, $Cu(II)$ oxidizes this radical to form a carbocation

Due to the greater steric hindrance above this carbocation compared to below, it is selectively captured from below by 1,1-diphenylethylene



[H][C@]1(CC#N)C=CC[C@H]1C[C+](C2=CC=CC=C2)C3=CC=CC=C3

CHECKPOINT

1 PTS

Due to the greater steric hindrance above this carbocation compared to below, it is selectively captured from below by 1,1-diphenylethylene

Finally, a proton is eliminated to form product A